Experiment 6: Time Series Analysis of Website Traffic

Aim:

To analyze the time series data of website traffic to identify trends, seasonality,

and anomalies.

Dataset:

A dataset containing daily website traffic data with columns such as Date, Page_Views, Unique_Visitors, and Bounce_Rate**.**

Methodology:

1. Load and Clean Data:

o Import the dataset and parse the Date column as a datetime object.

o Handle missing values by interpolation.

2. Time Series Decomposition:

o Decompose the time series data of Page_Views to identify trend, seasonality,

and residuals.

3. Visualization:

o Plot the time series data for Page_Views, Unique_Visitors, and Bounce_Rate.

o Use moving averages to smooth the time series and highlight trends.

o Create seasonal plots to visualize patterns over time.

4. Anomaly Detection:

o Identify and visualize anomalies in the time series data using statistical

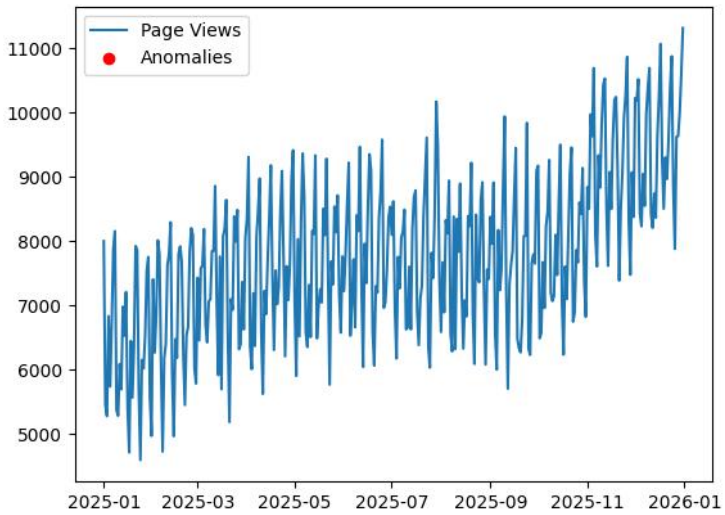
methods such as the Z-score.

Sample code:

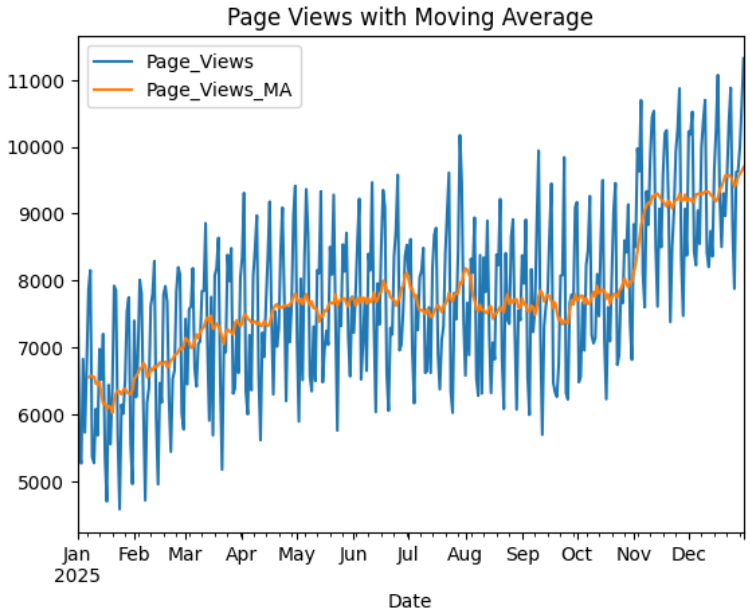
```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
data = pd.read_csv("website_traffic (1).csv",parse_dates=['Date'],index_col='Date')
decomposition = seasonal_decompose(
  data['Page_Views'],
  model='additive',
  period=7
)
decomposition.plot()
plt.show()
data['Page_Views_MA'] = data['Page_Views'].rolling(window=7).mean()
```

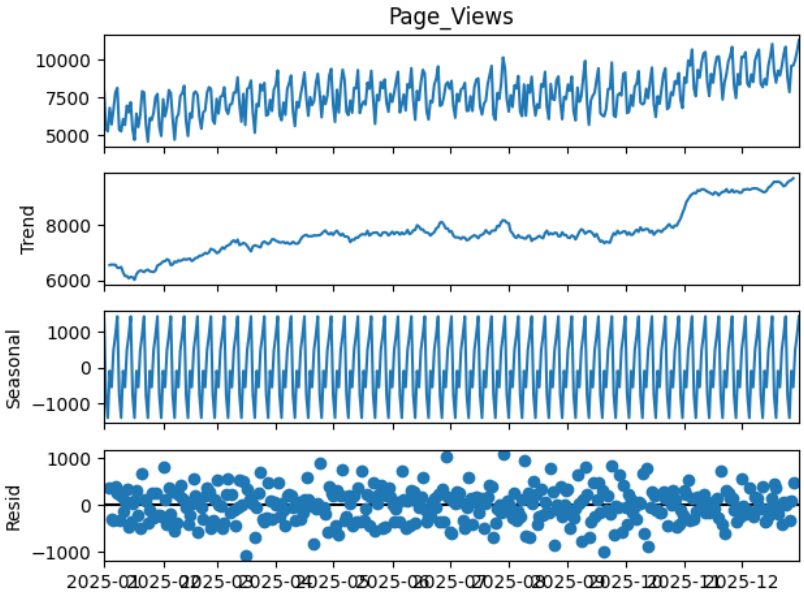
```
data[['Page_Views', 'Page_Views_MA']].plot()
plt.title('Page Views with Moving Average')
plt.show()
data['Page_Views_Z'] = (
  (data['Page_Views'] - data['Page_Views'].mean())
  / data['Page_Views'].std()
)
anomalies = data[data['Page_Views_Z'].abs() > 3]
plt.plot(
  data.index,
  data['Page_Views'],
  label='Page Views'
)

plt.scatter(
  anomalies.index,
  anomalies['Page_Views'],
  color='red',
  label='Anomalies'
)
plt.legend()
```




```
plt.show()
```





Result :

Thus Analysis of the time series data of website traffic has identified trends , seasonality and anomalies.